

MOBILE PROGRAMMING
LAB 3

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Create github Lab3 with Lab3 folder on computer. Submit assignments to the moodle system.

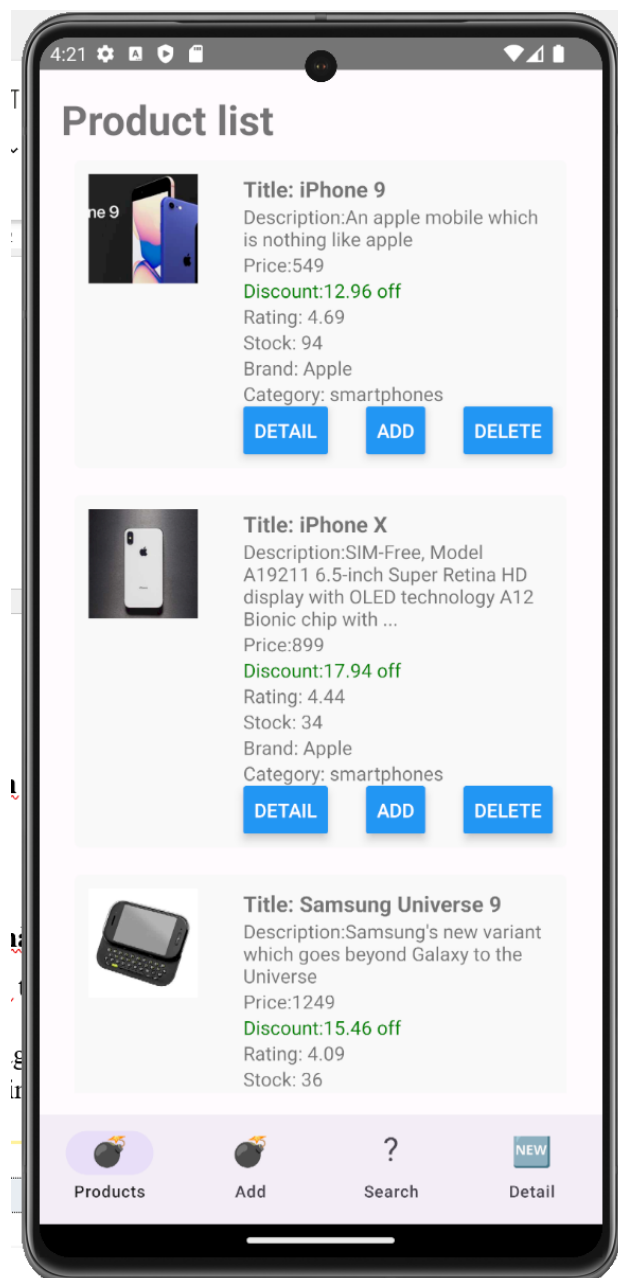
Create project Lab3, Create a Products folder

Install libraries:

npm install react-native-safe-area-context

npm install react-native-paper

Question 1. Develop the ProductList screen as below.



Note: Detail, add, delete buttons do not handle.

1. Create Products.js file
2. Use Flatlist to present the list.
3. The source code that reads the Products data returns Json data.

```
const [data, setData] = useState([])
const filePath = 'https://dummyjson.com/products/';
useEffect(() => {
  // Alert.alert(filePath);
  fetch(filePath)
    .then((response) => {
      if (!response.ok) {
        throw new Error('Network response was not ok');
      }
      return response.json();
    })
    .then((d) => {
      setData(d.products);
    })
    .catch((error) => {
      console.error('Error fetching data:', error);
    });
});
```

Question 2. Develop the “Add a Product” screen as below.

4:26

Add a Product

Title
Enter title

Description
Enter description

Price
Enter price

Discount Percentage
Enter discount percentage

Rating
Enter rating

Stock
Enter stock

Brand
Enter brand

Category
Enter category

Images
Enter images URL(s)

SUBMIT

Products Add Search Detail

1. Create Product_Add.js file
2. The source code adds data when the user clicks submit.

```

const [title, setTitle] = useState('');
const [description, setDescription] = useState('');
const [price, setPrice] = useState('');
const [discountPercentage, setDiscountPercentage] = useState('');
const [rating, setRating] = useState('');
const [stock, setStock] = useState('');
const [brand, setBrand] = useState('');
const [category, setCategory] = useState('');
const [images, setImages] = useState('');
handleSubmit = () => {
  fetch('https://dummyjson.com/products/add', {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({
      title: title,
      description: description,
      price: price,
      discountPercentage: discountPercentage,
      rating: rating,
      stock: stock,
      brand: brand,
      category: category,
      images: images,
    }),
  })
    .then((res) => res.json())
    .then(console.log);
  Alert.alert("Add sucessfull")
};

```

Question 3. Develop the Search products screen as below.



1. Create Product_Search.js file
2. Using CARD in react-native-paper library, Flatlist
3. The source code when the user clicks Search.

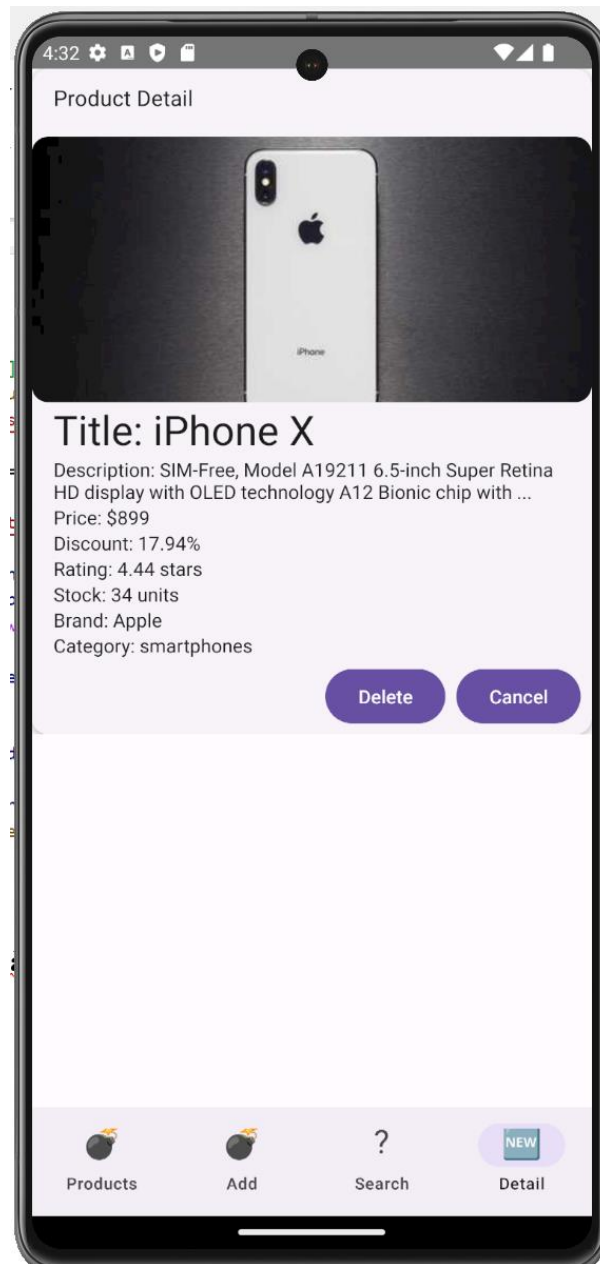
```

const [data, setData] = useState([]);
const [value, setValue] = useState('');
let filePath = 'https://dummyjson.com/products';

const searchProduct = () => {
  if(value!=='')
    filePath = 'https://dummyjson.com/products/search?q=' + value;
  fetch(filePath)
    .then((response) => {
      if (!response.ok) {
        throw new Error('Network response was not ok');
      }
      return response.json();
    })
    .then((d) => {
      setData(d.products)
    })
    .catch((error) => {
      console.error('Error fetching data:', error);
    });
};

```

Question 4. Develop the Detail screen as below.



Note: Do not handle delete, cancel buttons

1. Create `Product_Detail.js` file
2. Use `CARD`, `button` in `react-native-paper` library
3. Source code when a user retrieves a product

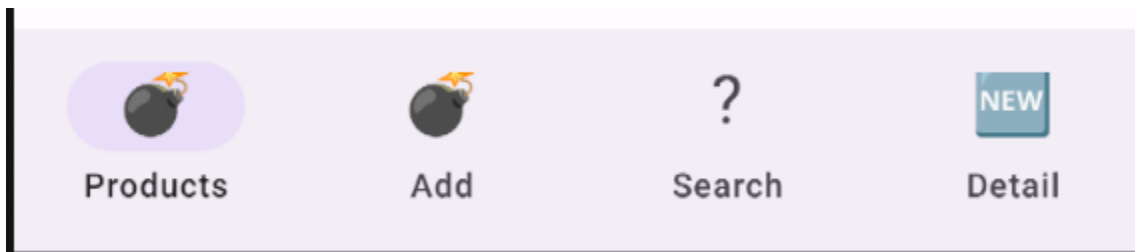

```

const [data, setData] = useState([])
const filePath = 'https://dummyjson.com/products/2';

useEffect(() => {
  // Alert.alert(filePath);
  fetch(filePath)
    .then((response) => {
      if (!response.ok) {
        throw new Error('Network response was not ok');
      }
      return response.json();
    })
    .then((d) => {
      setData(d);
    })
    .catch((error) => {
      console.error('Error fetching data:', error);
    });
});

```

Question 5. Develop Navigation as shown below.



1. Choose your own icon.
2. Use BottomNavigation in react-native-paper library, SafeAreaProvider in react-native-safe-area-context library.
Bottom navigation: <https://callstack.github.io/react-native-paper/docs/components/BottomNavigation/>
Icon: <https://callstack.github.io/react-native-paper/docs/guides/icons/>
3. Source code in App.tsx file.

```

import React, { useState } from 'react'

import ProductList from './Products/Products';
import Product_Add from './Products/Product_Add';
import ProductDetail from './Products/Product_Detail';
import ProductSearch from './Products/Product_Search';
import { BottomNavigation, Text } from 'react-native-paper'
import { SafeAreaProvider } from 'react-native-safe-area-context';

export default App = () => {
  const [index, setIndex] = useState(0);
  const [routes] = useState([
    { key: 'ProductList', title: 'Products', focusedIcon: 'folder'},
    { key: 'Product_Add', title: 'Add', focusedIcon: 'folder' },
    { key: 'ProductSearch', title: 'Search', focusedIcon: 'find' },
    { key: 'Product_Detail', title: 'Detail', focusedIcon: 'calendar' },
  ]);

  const renderScene = BottomNavigation.SceneMap({
    ProductList: ProductList,
    Product_Add: Product_Add,
    ProductSearch: ProductSearch,
    Product_Detail: ProductDetail,
  });

  return (
    <SafeAreaProvider>
      <BottomNavigation
        navigationState={{ index, routes }}
        onIndexChange={setIndex}
        renderScene={renderScene}
      />
    </SafeAreaProvider>
  );
};

```